

SSA3 (P09435) — Plasma Membrane

GO:0005886 Annotation Review

Hypothesis under review: The existing SSA3 IBA annotation to *plasma membrane* (GO:0005886) should receive review action **KEEP_AS_NON_CORE**.

Focus **type:** existing_go_annotation_decision · **Source:**
genes/yeast/SSA3/SSA3-ai-review.yaml → existing_annotations[3] · **Organism:**
Saccharomyces cerevisiae (NCBITaxon:559292)

Summary

The seed action **KEEP_AS_NON_CORE** is well justified in spirit: the plasma membrane is **not** a primary or core site of Ssa3 function. Every line of curated and experimental evidence localizes Ssa3 — a stress-inducible cytosolic Hsp70 of the Ssa family — to the **cytosol/cytoplasm**. The plasma-membrane assignment is a **phylogenetic (IBA) inference only**, carried on GO_REF:0000033, with no yeast-specific experimental support for Ssa3. Ssa3 also has **no transmembrane domain, signal peptide, or lipidation site**, so it possesses no intrinsic mechanism to reside in or anchor to a membrane.

One material fact tilts the practical decision toward **REMOVE** rather than merely keeping the term as non-core: a live check of GO_Central via QuickGO shows that the plasma-membrane annotation for P09435 **no longer exists** (GO:0005886 on P09435 → 0 hits), even though the UniProt cross-reference dump still lists it as a synchronization-lag artifact. Because the authoritative source has already retired this IBA, "keep as non-core" retains an annotation that upstream curation has already dropped. Both **KEEP_AS_NON_CORE** and **REMOVE** are curatorially reasonable and agree on the essential biology; **REMOVE** is now better aligned with the source of truth and with the paralog-over-annotation mechanism that produced the term.

A paralog audit explains where the family-level plasma-membrane signal actually comes from. The constitutive paralogs **Ssa1** and **Ssa2** were detected in a whole-cell *S. cerevisiae* plasma-membrane proteome (HDA, Delom et al. 2006, PMID:16622836) — a low-specificity context in which highly abundant cytosolic Hsp70 readily co-purifies with a membrane fraction. Ssa3 was **not** detected and now carries no PM annotation. Combined with mammalian Hsp70 orthologs

that have genuine experimental cell-surface localization, this is the ancestral-node source from which the (now-retired) IBA leaked onto the low-abundance, stress-inducible Ssa3/Ssa4. The verdict is therefore **partially supported**: the biological conclusion is correct, but the chosen action is weaker than the evidence warrants.

Executive Judgment

Verdict: Partially supported (KEEP_AS_NON_CORE is a defensible conservative action; REMOVE is the stronger, evidence-aligned alternative).

The core claim of the seed hypothesis — that plasma membrane is not a primary site of Ssa3 action — is **correct and well supported**. The weakness lies in the chosen mechanism: retaining an IBA that (1) has no Ssa3-specific experimental support, (2) is a paralog/cross-organism over-propagation artifact, and (3) has already been retired at GO_Central. The most important caveat is that this is a **localization nuance, not a functional error**, and that transient/peripheral membrane association by a cytosolic Hsp70 during client handling cannot be formally excluded — it is simply undocumented for Ssa3.

Paralog audit (QuickGO live, GO:0005886)

Protein	Accession	Expression	GO:0005886 status	Evidence / source
Ssa1	P10591	constitutive	present	HDA (ECO:0007005), P 16622836 (Delom 2006, PM proteome), SGD
Ssa2	P10592	constitutive	present	HDA (ECO:0007005), P 16622836 , SGD
Ssa3	P09435	stress-inducible	absent (0 hits)	IBA retired at GO_Central
Ssa4	P22202	stress-inducible	absent (0 hits)	—
HSC70/ HSPA8 (human)	P11142	—	present	IEA + TAS (Reactome)

Interpretation: the ancestral Hsp70 node inherits PM from Ssa1/Ssa2 (HDA) and mammalian Hsp70s, which is how the now-retired **IBA** leaked onto the low-abundance, stress-inducible Ssa3/Ssa4 — neither of which was detected in the PM proteome.

Key Findings

F001 — The plasma-membrane annotation is IBA-only, unsupported by yeast experimental evidence, and already retired at GO_Central

A direct audit of UniProt and QuickGO records for P09435 shows the plasma-membrane assignment never had experimental backing for Ssa3. UniProt's curated **SUBCELLULAR LOCATION** comment lists **Cytoplasm only**, and the keyword set includes **Cytoplasm** with **no membrane keyword**. The cellular-component GO annotations in UniProt were: GO:0005737 cytoplasm (IBA), GO:0005829 cytosol (**IDA:SGD**,

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— the only experimental CC record), GO:0005634 nucleus (IBA), and GO:0005886 plasma membrane (**IBA:GO_Central**).

When queried against the live QuickGO annotation service, P09435 returns five current CC-aspect records: cytoplasm (IBA), cytosol (IBA), nucleus (IBA), cytoplasm (IEA, GO_REF:0000120), and cytosol (IDA, ECO:0000314,).

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A targeted query for **GO:0005886 on P09435 returns** `numberOfHits = 0` — the plasma-membrane IBA is **no longer present** in GO_Central. The term originated purely as a phylogenetic inference whose ancestral `withFrom` set includes mammalian Hsp70s carrying experimental cell-surface / PM localization — HSC70/HSPA8 (P11142), HSPA1A/B (P0DMV8 / P0DMV9), HSPA6 (P17066), and HSPA1L (P34931). None of these are direct evidence for the yeast Ssa3 protein. This is the single most decision-relevant fact for the curator: the annotation being "kept" may no longer exist upstream.

F002 — The Ssa-family plasma-membrane signal traces to one PM proteomics study on the constitutive paralogs Ssa1/Ssa2, not Ssa3

To identify the true origin of the family-level signal, each Ssa paralog was audited in QuickGO for GO:0005886 (see paralog table above). Only Ssa1 (P10591) and Ssa2 (P10592) carry the term, each with a single **HDA** (ECO:0007005) record from

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(Delom, Szponarski, Sommerer, Boyer, *Proteomics* 2006, "*The plasma membrane proteome of *Saccharomyces cerevisiae* and its response to the antifungal calcofluor*"). Ssa3 and Ssa4 return **0 hits**. The two annotated paralogs are among the most abundant cytosolic proteins in yeast, and their detection in a plasma-membrane-enriched fraction is readily explained as low-specificity co-purification rather than bona fide PM residence. Human HSC70/HSPA8 additionally carries GO:0005886 via IEA and TAS (Reactome). The Ssa3 phylogenetic ancestral node would inherit plasma membrane from the Ssa1/Ssa2 HDA evidence **and** from the mammalian Hsp70s, producing the (now-retired) IBA on Ssa3/Ssa4 despite no direct observation. This is a textbook **paralog over-annotation** pattern.

F003 — Ssa3 has no membrane-targeting sequence features; its sole experimental localization is cytosolic

An independent sequence-level check confirms no biophysical basis for a plasma-membrane residence. A UniProt feature audit of P09435 returns feature types limited to {Chain:1, Region:1, Compositional bias:1, Sequence conflict:1} — with **no transmembrane segment, no signal peptide, and no lipidation site**. Consistent with this, the only experimental CC annotation for Ssa3 (GO:0005829 cytosol, IDA, ECO:0000314) derives from

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(Satyanarayana, Schröder-Köhne, Craig, Schu, *FEBS Letters* 2000, "Cytosolic Hsp70s are involved in the transport of aminopeptidase 1 from the cytoplasm into the vacuole"), a study that explicitly treats the Ssa Hsp70s as cytosolic chaperones. Absence of membrane-targeting features plus positive cytosolic evidence closes the loop: any plasma-membrane association, if it occurs, would be transient/peripheral, not a stable localization or a dedicated site of function.

Mechanistic Model / Interpretation

EXPERIMENTAL REALITY (yeast)

Ssa1/Ssa2 (constitutive, abundant)

high cytosolic abundance
co-fractionates w/ PM prep

Mammalian HSC70/HSPA1 etc. —exp→

ANNOTATION PROPAGATION PATH

GO:0005886 PM (<https://pubmed.ncbi.nlm.nih.gov/10745074/>)

phylogenetic tree
▼ (IBA, GO_REF:0000033)

Ancestral Hsp70 node inherits PM
(also from mammalian cell-surface Hsp70s)

▼ over-propagation

Ssa3 (P09435) IBA GO:0005886 ← REVIEW TARGET

▼ GO_Central re-curation

RETIRED (QuickGO: 0 hits today)

Ssa3 DIRECT EVIDENCE:

- Cytosol IDA (<https://pubmed.ncbi.nlm.nih.gov/10745074/> rel="noopener noreferrer")
- Cytoplasm (UniProt curated comment)
- No TM / signal peptide / lipidation ← no membrane-targeting features

Ssa3 is a stress-inducible cytosolic Hsp70 whose documented molecular activities —
 chaperoning Hsp90 clients
 ([P 32299842](#)),
 regulating ribonucleotide reductase via its C-terminal domain
 ([P 35417483](#)),
 promoting autophagic degradation of toxic aggregates
 ([P 30376576](#)),
 and prion propagation / thermotolerance
 ([P 24628813](#))
 — are all **cytosolic/nuclear**. None require or imply a plasma-membrane site of action. The
 plasma-membrane term is therefore best understood as an annotation-provenance artifact
 rather than a reflection of Ssa3 cell biology.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con limi
UniProt P09435 (curated CC comment)	Review/ database	Refutes PM as core	Where is Ssa3 localized?	Curated SUBCELLULAR LOCATION = Cytoplasm only; keyword "Cytoplasm"; no membrane keyword	<i>S. cerevisiae</i>	HIGH cyto leve
PMID:10745074 (Satyanarayana 2000; SGD IDA, GO:0005829)	Localization (direct assay)	Supports cytosol; refutes PM as primary	Experimental localization	Only experimental CC annotation = cytosol (ECO:0000314, IDA); "Cytosolic Hsp70s... transport of aminopeptidase 1 from the cytoplasm into the vacuole"	<i>S. cerevisiae</i>	Mod sing expe CC r
UniProt P09435 sequence features	Structural/ sequence	Refutes PM residence mechanism	Any membrane- anchoring signal?	No transmembrane, signal-peptide, or lipidation features — soluble protein with no PM- targeting mechanism	<i>S. cerevisiae</i>	HIGH of-fe argu
GO_Central IBA GO:0005886 (GO_REF:0000033)	Computational (phylogenetic)	The annotation being reviewed	Is PM directly evidenced?	PM assigned by IBA only ; ancestral withFrom includes	Phylogenetic node	Low infe yeas

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con limi
				mammalian Hsp70s (P11142; P0DMV8/9; P17066; P34931) with experimental cell-surface localization		
QuickGO live query (2026-08)	Database status	Qualifies/ strengthens removal	Does PM IBA still exist?	Direct query P09435 + GO:0005886 → 0 hits ; annotation retired at GO_Central	GO_Central	High check
PMID:16622836 (Delom 2006; HDA on Ssa1 & Ssa2)	Localization (HT proteomics)	Qualifies (family- level, not Ssa3)	Any yeast PM evidence for the Ssa family?	Constitutive Ssa1/Ssa2 detected in PM proteome; Ssa3 not detected	<i>S. cerevisiae</i> PM fraction	Low spec abun chap men prep cont prom
PMID:24628813 (Hasin 2014)	Mutant phenotype / expression	Supports cytosolic function	Ssa3 functional role	Ssa1–4 are "cytosolic Hsp70-Ssa family"; Ssa3 stress-inducible, most proficient for [PSI ⁺] prion	<i>S. cerevisiae</i>	High cont
PMID:35417483 (Knighton 2022)	Mutant phenotype / interaction	Supports cytosolic function	Ssa3 role & identity	"four highly similar cytosolic Hsp70s	<i>S. cerevisiae</i>	High

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Con limi
				Ssa1,2,3,4... Ssa3 and 4 induced upon heat shock"; regulate RNR		
PMID:32299842 (Gaur 2020)	Interaction / phenotype	Supports cytosolic function	Ssa3 in Hsp90 pathway	Ssa3 acts as cytosolic Hsp70 modulating Hsp90 client (v- Src) maturation	<i>S. cerevisiae</i>	High
PMID:30376576 (Gupta 2018)	Mutant phenotype	Supports cytosolic function	Ssa3 protective role	Stress-inducible Ssa3/4 reduce α - synuclein toxicity via autophagy; treated as cytosolic Hsp70	<i>S. cerevisiae</i>	High

Evidence Base (literature synthesis)

The functional literature is unanimous that Ssa3 is a **cytosolic** Hsp70 and never invokes a plasma-membrane role. [PMID:35417483](#) states directly that budding yeast "express four highly similar cytosolic Hsp70s Ssa1, 2, 3 and 4," with Ssa3/4 induced on heat shock, and assigns Ssa3 a role in regulating ribonucleotide reductase (a cytosolic/nuclear process) via its C-terminal domain. [PMID:24628813](#) characterizes the Ssa isoforms as "the cytosolic Hsp70-Ssa family" and finds Ssa3 most proficient for [PSI⁺] prion propagation — again a cytosolic function. [PMID:32299842](#) places Ssa3 in the cytosolic Hsp90 client-maturation pathway (v-Src), and [PMID:30376576](#) shows stress-inducible Ssa3/4 reduce α -synuclein toxicity via autophagy. The only two localization-relevant primary sources are [PMID:10745074](#) (Ssa3 cytosol, IDA —

supports the core CC) and [PMID:16622836](#) (PM proteome detecting only Ssa1/Ssa2 — the qualifying, family-level source of the propagated PM term). No primary study documents Ssa3 at the plasma membrane.

GO Curation Implications

- **Aspect:** Cellular Component (CC). Term = plasma membrane (GO:0005886), evidence IBA, ref GO_REF:0000033.
- **Primary lead** — **REMOVE** the plasma-membrane IBA for SSA3. Rationale: (1) no yeast experimental support; (2) contradicted in emphasis by the only experimental CC annotation (cytosol, IDA,

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and the UniProt curated location (Cytoplasm); (3) it is an ancestral-node over-propagation from mammalian Hsp70 paralogs plus the constitutive Ssa1/Ssa2 HDA; (4) **GO_Central has already retired it** (QuickGO returns 0 hits), so removal restores consistency with the authoritative source.

- **Acceptable fallback** — **KEEP_AS_NON_CORE** (the seed action) if the curator prefers to preserve a low-confidence peripheral-association possibility. This is defensible but effectively re-instates an annotation the source has dropped.
- **Retain as core:** GO:0005829 cytosol (IDA) and GO:0005737 cytoplasm — these are the supported CC terms. Do **not** upgrade or make PM more specific; there is no evidence for a specific membrane subdomain.

GO decision table (leads — require curator verification)

GO term	Aspect	Current evidence for Ssa3	Recommended action	Confidence
GO:0005886 plasma membrane	CC	IBA only (retired at GO_Central; 0 hits); no Ssa3 experimental data; no TM/signal/lipid	REMOVE (or KEEP_AS_NON_CORE as conservative fallback)	High that PM is non-core
GO:0005829 cytosol	CC	IDA (P 10745074 ECO:0000314) + IBA	RETAIN as core	High
GO:0005737 cytoplasm	CC	IBA + IEA; curated UniProt location	Retain (parent of cytosol)	High
GO:0005634 nucleus	CC	IBA	Retain as non-core (plausible for Hsp70 clients/DDR)	Moderate

Mechanistic Scope

Ssa3 has **no transmembrane domain, signal peptide, or lipidation site** in UniProt (P09435) — i.e., no intrinsic mechanism to reside in or anchor to the plasma membrane, reinforcing that any PM detection is peripheral/adventitious. The immediate molecular function of Ssa3 is ATP-dependent Hsp70 chaperone activity in the **cytosol** (holdase/foldase for misfolded and nascent clients; partner to Hsp90 and J-domain co-chaperones Ydj1/Sis1; nucleotide-exchange-regulated client binding/release). Its documented cellular processes — prion propagation, protein-aggregate/ α -synuclein clearance, RNR regulation, Hsp90 client maturation, thermotolerance — all occur in the cytosol/nucleus. No direct membrane-anchoring, transmembrane, or lipid-binding activity is annotated. Any plasma-membrane presence would be a **transient, peripheral** consequence of client handling or translocation, i.e., downstream of the core cytosolic chaperone role — not an intrinsic localization.

Conflicts and Alternatives

- **Paralog over-annotation (most likely explanation):** The Hsp70 ancestral node inherits plasma membrane from two sources — the yeast constitutive paralogs Ssa1/Ssa2 (HDA,

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and mammalian Hsp70s with experimental cell-surface localization. Phylogenetic propagation carried "plasma membrane" onto the stress-inducible Ssa3/Ssa4 as IBA, even though neither was detected in the yeast PM proteome. GO_Central has since retired the Ssa3/Ssa4 IBA while retaining the HDA calls on Ssa1/Ssa2.

- **High-throughput artifact (why even the Ssa1/Ssa2 HDA is weak):** Ssa1/Ssa2 are among the most abundant cytosolic proteins; their appearance in a plasma-membrane preparation is a classic low-specificity co-purification and does not establish bona fide PM residence, further undermining any inference that Ssa3 functions at the PM.
- **Database carry-over:** UniProt still lists the GO:0005886 IBA cross-reference while GO_Central has dropped it — a synchronization lag, not new evidence. Curators should treat the QuickGO live status as authoritative.
- **Not excluded:** Genuine transient membrane association during co-translational translocation or client delivery is biologically plausible for cytosolic Hsp70s but is unproven for Ssa3 specifically.

Limitations and Knowledge Gaps

1. **Is there any yeast experimental PM/cell-surface evidence for Ssa3?** Checked UniProt, QuickGO, and PubMed — none found. Matters because it is the sole basis that could justify keeping the term. Resolve with a targeted GFP/immuno-EM or surface-biotinylation study of tagged Ssa3.

2. **Identity/robustness of the cytosol IDA**

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Verified it exists as the single experimental CC record; did not read the full paper. Matters for weighting cytosol as core. Resolve by confirming the assay (likely fractionation/localization).

3. **Exact GO_Central removal history.** QuickGO shows 0 hits now; the removal date/reason was not retrieved. Matters for justifying REMOVE vs KEEP. Resolve via GO annotation change logs.
4. **Delom 2006 peptide-level specificity.** Because Ssa paralogs share extensive sequence identity, I did not confirm from the raw peptide tables whether Ssa1/Ssa2 assignments rest on paralog-discriminating peptides. This bears on how confidently Ssa3 can be called "absent" from the PM proteome.

Discriminating Tests / Proposed Follow-up Actions

- **Localization:** High-resolution live-cell imaging (endogenously tagged GFP-Ssa3) ± heat shock, with a PM marker (e.g., Pma1-mCherry), plus cell-surface biotinylation/flow cytometry to test for any surface pool. Expected: diffuse cytosolic signal, no PM enrichment.
- **Fractionation:** Sucrose-gradient / membrane-vs-cytosol fractionation of induced Ssa3 (heat shock), with protease-protection to distinguish peripheral from integral association. Expected: fully soluble/peripheral.
- **Comparative annotation audit:** Confirm whether Ssa1 (P10591), Ssa2 (P10592), and Ssa4 (P22202) still carry GO:0005886; consistent removal across the stress-inducible paralogs confirms the over-propagation interpretation.
- **Provenance check:** Compare the GO release used to build the AIGR review against current GO_Central to determine whether the action is "remove" or "confirm already-removed."

Curation Leads (require curator verification)

- **Candidate action change:** `KEEP_AS_NON_CORE` → `REMOVE` for GO:0005886 (IBA, GO_REF:0000033), justified by absence of yeast evidence and confirmed retirement at GO_Central. If the curator prefers maximum conservatism, retain the seed `KEEP_AS_NON_CORE` with an explicit note that the term is IBA-only and source-retired.
- **Candidate references / snippets to verify:**
- UniProt P09435: curated `SUBCELLULAR LOCATION: Cytoplasm`; CC GO includes `GO:0005829 cytosol` | `IDA:SGD` and `GO:0005886 plasma membrane` | `IBA:GO_Central`.

- QuickGO live: `GET annotation/search?geneProductId=P09435&goId=G0:0005886` → `numberOfHits = 0`.

- P 16622836

 - "*The plasma membrane proteome of Saccharomyces cerevisiae and its response to the antifungal calcofluor*" (supports only Ssa1/Ssa2 at the PM).
- P 10745074

 - "*Cytosolic Hsp70s are involved in the transport of aminopeptidase 1 from the cytoplasm into the vacuole*" (supports cytosol as core CC).
- P 35417483

 - "cells express four highly similar cytosolic Hsp70s Ssa1, 2, 3 and 4... Ssa3 and 4 are induced upon heat shock."
- **Suggested curator questions:** (1) Should source-retired IBA terms be removed rather than downgraded? (2) Is there any organism-specific PM evidence justifying retention?
- **Suggested experiment:** Surface-biotinylation + fractionation of induced GFP-Ssa3 to settle the peripheral-association question.

Analyses executed live against UniProt REST and EBI QuickGO on 2026-08-22; PubMed evidence as cited. Computational results reported conservatively; direct database queries distinguished from phylogenetic inference.
